

# Project

Danica Milosavljevic

June 2026

## 1 Introduction

### 1.1 Social dynamic crisis

The landscape of global mental health has shifted drastically over the past years. According to the recent statistics from World Health Organization, there are over one billion people that are living with recognised mental health condition. This makes mental illness one of the leading drivers of global disability. The increase of people with these issues is strongly correlated with rising problems in social dynamics, most notably the psychological effects of COVID-19 pandemics.

The post pandemic era had major influence on traditional communication patterns. Extended periods of isolation have created strong reliance on digital first interactions, which caused widespread social anxiety. For many people navigating face to face interactions became stressful task, driving away people from physical support and pushing them to unhealthy coping strategies.

### 1.2 Economic crisis

However, human crisis is not the only problem when dealing with global mental health problem. Globally, there is a huge mismatch between the demand for the psychological support and economic resources that are allocated for it.

Massive macroeconomic crisis happens with low productivity, absenteeism, premature economic exit, which are often consequences of mental health struggles. In addition, despite the fact that globally around 8% of medical problems are related to psychological issues, only 2% of the overall health budget is allocated to mental health care, which is almost unchanged for almost a decade.

The systematic underfunding leaves majority of people without accessible help. In high income countries waiting list for clinical psychological evaluations routinely goes from six months to over a year. In low-to-middle income nations, the ratio of professionals to citizens drops significantly to as few as one psychiatrist per million people. Private therapy remains an luxury, leaving the vast majority of suffering populations with zero options for structured support.

### 1.3 The Gender Disparity in Medical Research

In the current system, there is evident neglect in women’s mental health. Women are diagnosed with generalized anxiety disorder and major depressive disorders at twice the rate of men. Furthermore, younger women aged 16–24 are nearly three times as likely to experience an acute mental health issue as their male peers. Despite this clear epidemiological reality, medical and psychological research has historically suffered from a profound male bias, creating a structural data deficit that leaves women at a distinct disadvantage when seeking help. This data gap is driven by three main historical and structural factors:

- Systematic trial exclusion that was present until mid 1990s, that happened because of the assumption that hormones would create noisy data.
- Data exploration bias: Because primary datasets were constructed using male subjects, the clinical baselines for diagnosis and pharmaceutical treatments used today are largely extrapolated from male physiology and psychology.
- Critical endocrine milestones unique to female biology, such as the postpartum phase, the premenstrual matrix, and perimenopause are drastically under researched, frequently leaving patients with answers that overlook the intersection of hormonal changes and neurological wellness.

### 1.4 Benefits of RAG inclusion

As a consequence of clinical bottlenecks and research gaps, individuals are increasingly using generative AI architectures for support. However, standard Large Language Models (LLMs) present severe deployment risks in sensitive domains, frequently generating hallucinations, or delivering unverified medical advice. To resolve this safely, Retrieval-Augmented Generation (RAG) introduces a strict architectural constraint. Instead of allowing an LLM to generate text solely from its internal weights, a RAG system forces the model to act as educated dashboard of a verified, external reference library. When a user submits a query, the system first passes it to a semantic vector database containing curated, open-source wellness data. The system retrieves the most relevant text chunks and uses them as a contextual ground for the Large Language Model. The LLM then generates a response that is strictly bound by these documents, completely avoiding medical prescriptions or the simulation of personal relationships.

General benefits of using RAG architecture are minimising the risk of hallucinations that is very common in LLMs, traceable source citations and cost-efficiency compared to fine-tuning, that can cost thousands of dollars.

When applied to women’s mental health, a RAG framework allows a model to bypass the historical biases of mainstream medicine by building a curated, high-integrity knowledge base drawn directly from open-source lifestyle, nutritional, and psychoeducational literature. Crucially, because this system operates at the intersection of lifestyle adjustments, such as soundscapes, meditation preferences, somatic exercises, and endocrine-conscious nutritional adjustments, it sidesteps the clinical dangers of automated medical prescriptions. The archi-

ture ensures complete user anonymity and provides free access to wellbeing protocols.

## **1.5 Establishing educational non-relational boundary**

Fundamental part of this project is the complete rejection of the artificial or therapeutic personal relationships. Unlike conversational agents that attempt to simulate human empathy or act as artificial friends, this project is designed strictly for educational and knowledge-delivery purposes. The system treats the user not as a patient in a clinical relationship, but as a self-directed agent seeking high-integrity, evidence-based wellness data. By positioning the interface as an interactive reference dashboard rather than an intimate chat companion, the project eliminates the ethical hazard of misleading vulnerable users into believing they are conversing with an emotional entity. It focuses strictly on providing an objective opportunity for knowledge sharing and self-guided improvement.

## **1.6 Existing wellbeing chatbots**

Current leading paradigms in this digital mental health space are Woebot and Wysa. While both platforms gave successful access to psychological tools, there are gaps in operational and philosophical areas that RAG architecture aims to improve.

Historically, Woebot operates on a deterministic, rules based architecture rather than a fully generative one. While it utilizes Natural Language Processing (NLP) at the entry point, it uses this technology purely for input classification, mapping the user’s words to specific intent categories. Once the intent is classified, the system routes the user down a pre scripted, human-authored conversational branching path rooted in Cognitive Behavioral Therapy (CBT). The primary limitation of these fixed scripts is that because every response originates from a static folder of pre-approved statements, the interaction can quickly feel repetitive. If a user’s queries falls outside the pre-programmed tree, the illusion of understanding breaks down, leading to a mechanical and frustrating user experience that cannot adapt to highly individualized lifestyle queries. Wysa was traditionally built using a similar hybrid model, combining structured AI guided paths with human-in-the-loop coaching. In recent iterations, the platform has begun integrating clinical Large Language Models to handle complex user inputs and draft weekly reflection summaries. However, because it operates as a commercial, clinical-first application, its main design goal remains relational. It explicitly seeks to establish a persistent digital therapeutic alliance and companion-oriented ongoing check-ins to build emotional compliance.

Architecture of RAG represents a significant evolutionary step forward from both rigid rules-based chatbots and unconstrained commercial conversational agents. It successfully unifies the strengths of natural language expression with absolute factual safety across three distinct technical are: First, regarding con-

versational flexibility, rules-based systems like traditional Woebot are limited to static branching paths, and unconstrained commercial systems present massive safety risks. The RAG architecture offers a dynamic yet bounded approach, where natural language generation remains fluid but completely pinned to external texts. Second, regarding the user relationship goal, while existing market players seek to build continuous companion-oriented bonds or digital alliances, this project implements a strictly educational model. It functions purely as an objective, high-integrity knowledge-sharing portal. Third, regarding risk management, RAG completely mitigates the high user friction and robotic repetition found in rules-based systems, while simultaneously eliminating the data hallucinations, prompt injections, or accidental medical diagnoses common to standard generative configurations. By transitioning away from rigid, pre-scripted buttons and moving toward a RAG architecture, this project provides users with a fluid, natural language interface that can answer highly specific, granular wellness questions. Simultaneously, by sourcing its knowledge directly from an explicit database of open-source literature on nutrition, somatic exercise, and mindfulness, it guarantees that the output remains entirely factual, traceably cited, and strictly non-medical.

## **2 Related Work**

- Work that is connected to my thesis.
- Explanation of read research papers

## **3 Methodology**

- Describing method and terminology in detail
- RAG, Multi-Agent, Choice of Agents, Podcasts
- The benefits of using current method
- Results that are obtained

## **4 Results and Evaluation**

- Analysis of results
- Comparison of different approaches
- Limitations and extensions of current work

## **5 Conclusion**